MUSKAN

muskan97820@gmail.com | LinkedIn | GitHub | Portfolio | 9782005139

Education

Indian Institute of Technology Dhanbad — **Dec 2020 – May 2024**
B.Tech. in Mechanical Engineering (Spl. Mining Machinery) — *CGPA: 8.13*

Skills Summary

Programming: Python, SQL, C++, JavaScript
Data Engineering: ETL/ELT, Data Pipelines, Data Modeling, PySpark, Apache Airflow, Apache Kafka
Cloud: Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Synapse Analytics, GCP
Databases: MS SQL Server, MySQL, MongoDB, Firebase
Tools & Analytics: Git, Power BI, SAP, MS Excel, Postman, Machine Learning

Experience

Element14 — **June 2025 – Present**
Business Intelligence Analyst – I — *Bengaluru, Karnataka*
- Developed a Python/Pandas/RapidFuzz fuzzy-matching pipeline to identify reseller accounts across 6M+ records using token normalization, Levenshtein similarity and chunked processing; reduced manual matching for 90+ weekly records from 30+ hours to 5 minutes.
- Automated data extraction, transformation and distribution across Adobe APIs, SQL Server, Outlook and FTP for all regions, reducing preparation time for 5 monthly reports from 1 week to half a day while minimizing manual errors.
- Built a Python customer reactivation pipeline for 6.4M+ customers, engineering 30+ behavioral/transactional features and training a Random Forest model with temporal validation to predict 365-day reactivation.

Jindal Power Limited — **July 2024 – May 2025**
Graduate Engineer Trainee — *Raigarh, Chhattisgarh*
- Developed an ML-based predictive maintenance model using lubrication, power consumption, speed, temperature and breakdown data to identify failure risks and enable proactive maintenance before critical failure, helping prevent unplanned plant shutdowns.
- Redesigned a closed-loop coal handling circuit from crusher to final delivery, performing capacity analysis, equipment sizing, process optimization, feasibility and cost estimation for an INR 350+ Cr project selected for implementation.

Projects

End-to-End Azure Data Pipeline for E-Commerce Analytics — **GitHub Link**
ADF, Databricks, PySpark, ADLS Gen2, Synapse, MongoDB
- Built an Azure data pipeline using ADF to ingest 8 e-commerce datasets from GitHub/HTTP and MySQL into ADLS Gen2 Bronze using configuration-driven sequential ingestion and ARM templates.
- Implemented Medallion Architecture (Bronze-Silver-Gold) with Databricks/PySpark for deduplication, feature engineering, 5 sequential joins and MongoDB product-category enrichment.
- Developed Python ingestion workflows for MySQL/MongoDB using batch processing and bulk inserts; wrote consolidated datasets to the ADLS Gen2 Silver layer in Parquet.
- Built the Synapse serving layer using views and CETAS with Parquet/Snappy compression to materialize curated Gold datasets in ADLS Gen2 for Power BI analytics.

Real-Time Fraud Detection & Risk Engine — **GitHub Link**
Kafka, PySpark Streaming, XGBoost, Delta Lake, Airflow, Docker, GCP
- Built an event-driven fraud pipeline using Kafka and PySpark Streaming, applying event-time windows, watermarks, stateful processing and stream joins to transaction, login, payment and location events.
- Engineered rule-based fraud detection with XGBoost risk scoring across velocity, impossible travel, login correlation, payment attacks and high-value anomalies; implemented Delta Lake Bronze-Silver-Gold layers with Airflow-driven training, quality checks and backfills.

Honors and Awards

- Awarded **Spot Awards** in two quarters at Element14 for contributions supporting the team.
- Received the **Extra Mile Award** for supporting cross-functional teams beyond core responsibilities.
- Solved 500+ problems across GeeksForGeeks, CodingNinjas, AtCoder, Codeforces and HackerEarth.